

Literature Status: Monotonicity of FL^p and $a-FL^p$

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Status

This is a literature-already-answered packet, not an original proof from the run. The first displayed question in Section “Current developments and open questions” of Chatterji–Drutu–Haglund, arXiv:0704.3749, asks whether, for every $p \geq q \geq 2$,

$$FL^p \implies FL^q \quad \text{and} \quad a-FL^q \implies a-FL^p.$$

Marrakchi–de la Salle, arXiv:2001.02490, answer this question affirmatively.

Original Question

The original source is:

I. Chatterji, C. Drutu, and F. Haglund, *Kazhdan and Haagerup properties from the median viewpoint*, arXiv:0704.3749.

Definition 1.4 says that a topological group G has property FL^p when every continuous affine isometric action of G on a space $L^p(X, \mu)$ has bounded orbits, and that G is $a-FL^p$ -menable when it has a proper continuous affine isometric action on some $L^p(X, \mu)$.

The first question in Section “Current developments and open questions” asks whether Corollary 3 generalizes to the monotonicity statement above.

Later Answer

The supporting source is:

A. Marrakchi and M. de la Salle, *Isometric actions on L_p -spaces: dependence on the value of p* , arXiv:2001.02490.

Its abstract explicitly says that it answers a question by Chatterji–Drutu–Haglund. Theorem 1 states that if $0 < p \leq q < \infty$, then every continuous affine isometric action $\alpha : G \curvearrowright L_p$ gives a continuous affine isometric action $\beta : G \curvearrowright L_q$ such that

$$\|\alpha_g(0)\|_{L_p}^p = \|\beta_g(0)\|_{L_q}^q \quad (g \in G).$$

Consequently unbounded orbits, and metrically proper orbits, pass from exponent p to every larger exponent q .

Corollary 2 records this in interval form: for every topological group G , the exponents admitting a continuous affine isometric L_p action with unbounded orbits form (p_G, ∞) or $[p_G, \infty)$, and the exponents admitting a proper continuous affine isometric L_p action form (p'_G, ∞) or $[p'_G, \infty)$.

Translation

The negation of FL^r is the existence of a continuous affine isometric L^r action with unbounded orbits. Since the set of such exponents is an upper interval, its complement is downward closed. Hence if $p \geq q \geq 2$ and G has FL^p , then G has FL^q .

Similarly, $a\text{-}FL^r$ -menability is exactly the existence of a proper continuous affine isometric action on some L^r space. Since the proper action exponents form an upper interval, $a\text{-}FL^q$ -menability implies $a\text{-}FL^p$ -menability for every $p \geq q \geq 2$.

Verification Notes

This is not a same-paper extraction false positive: arXiv:0704.3749 asks the question in 2007/2009, while arXiv:2001.02490 is a later independent paper. The supporting paper explicitly identifies the Chatterji–Drutu–Haglund question and says that Corollary 2 answers it. Its introduction also states that throughout the paper an L_p space means $L_p(X, \mu)$ for a standard measure space, matching the formulation of FL^p in the original source.

Limitations

This packet only records the literature answer to the monotonicity question. It does not compute the constants p_G or p'_G , and it does not settle the separate question in arXiv:0704.3749 about equivalence of different large FL^p properties or prescribed threshold examples.

References

- [1] I. Chatterji, C. Drutu, and F. Haglund, *Kazhdan and Haagerup properties from the median viewpoint*, arXiv:0704.3749.
- [2] A. Marrakchi and M. de la Salle, *Isometric actions on L_p -spaces: dependence on the value of p* , arXiv:2001.02490.